

REAL

Television

Operating Instructions &

Warranty

1413558-2
AW518

Important Information Concerning Your Warranty and Consumer Product Safety Act

There are two important reasons for the prompt warranty registration of your new RCA television set. The first is to assure that you immediately receive the full benefit of the parts and service labor warranties applicable to your purchase. The second, and equally important reason, is to assure that you will promptly receive any notification of inspection, modification, or recall that RCA may be required to give under the 1972 Consumer Product Safety Act.

The dealer, from whom you purchased your RCA television set, should have registered the set and given you the "Customer Copy" of the registration form. It looks like this:

RCA Warranty Registration

Model Number: A4097Y
 Date of Sale: 526213001
 Customer: John Doe
 Address: 560 Thomas St.
 City: Harrisburg, PA 17111
 State: PA
 Zip: 17111

RCA Consumer Electronics

Customer Copy

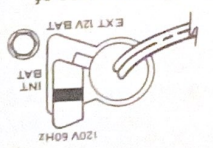
If you did not receive your copy of this form, you should IMMEDIATELY ask your dealer to promptly provide you with it.

If your dealer fails to provide you with a copy of the form, you may write to RCA at the address in the warranty statement included herein, enclosing a copy of your sales slip and necessary information (such as Model Number and Serial Number) if it does not already appear on the sales slip. Upon receipt of this information, RCA will register your warranty and return the Customer Copy to you.

Warning: To prevent fire or shock hazard, do not expose this TV to rain or moisture.

This TV has been engineered to provide adequate ventilation for cool operation. Do not place the TV where air circulation is restricted at the bottom, top or back. Cleaning the ventilation slots on the cabinet at regular intervals will assure adequate ventilation. To clean the slots, unplug the set and use the crevice attachment on your vacuum cleaner. (Do not use liquids or any object that will go thru the slots.)

Operating the TV on 120 Volt House Electrical System

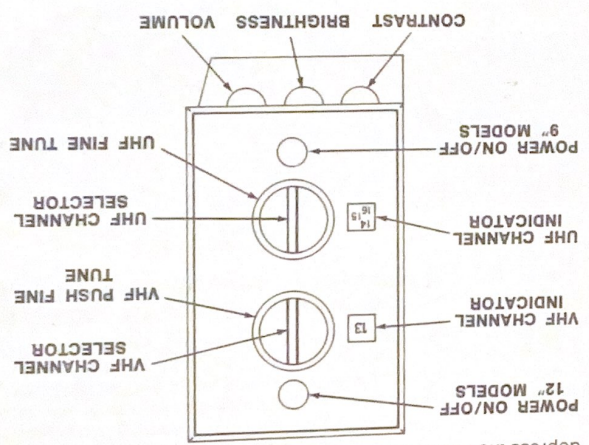


Operate the TV on 120 volt house electrical system first by placing the TV in the "120V 60 Hz" position. Plug the AC power cord into the TV and the wall outlet.

The functions of all controls are described in sequence of complete initial tuning. After this initial tuning, the controls will require little readjustment.

Important: Read "Charging Batteries" before operating the TV on internal battery.

To switch the TV "on" depress the Power button; the picture will appear in about six seconds. To switch the TV "off" depress the Power button again.



VHF Tuning (Channels 2 to 13)

Select desired VHF channel by rotating the VHF Channel Selector knob until desired channel number is indicated.

Adjust volume by rotating the volume knob (clockwise for increase in volume) to the desired level.

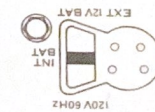
Fine Tune for best picture quality. Push in and turn the VHF Fine Tune knob clockwise until harsh interference occurs and the picture begins to disappear. Then turn slowly back, stopping as soon as all interference disappears and

When to Charge: The picture will begin to lose width after operating the TV approximately $3\frac{1}{2}$ to 4 hours on battery power. For maximum battery life, discontinue on battery operation when the picture pulls in $\frac{1}{4}$ inch on either side. Recharge the batteries 18 to 24 hours.

Operating TV on Internal Battery

Disconnect the AC cord from the wall outlet and from the TV. Place AC/DC switch located on the rear of the TV in the "INT BAT" position.

Depress the Power button and operate the TV in the same manner as when using AC power.



Viewing time with fully charged batteries is approximately $3\frac{1}{2}$ to 4 hours.

Battery Storage

Batteries should be fully charged before storage and recharged every six months. If stored in temperatures above 77°F , batteries require more frequent recharging.

Caution

In the unlikely event the battery seal should become damaged, leakage can occur, which might be injurious to you.

These are recommended antidotes:

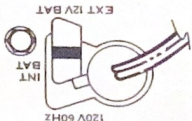
External—Flush with water.

Internal—Drink large quantities of water or milk. Follow with Milk of Magnesia, beaten eggs, or vegetable oil. Call a physician immediately.

Eye—Flush with water for 15 minutes and get prompt attention.

Operating TV on External Battery

The TV is designed to operate from a 12 volt DC automobile electrical system which has a negative ground. Use the DC power cord designed to plug into an automobile cigarette lighter receptacle.



Disconnect the AC cord from the wall outlet and from the TV. Place the AC/DC switch located on the rear of the TV in the "EXT 12V BAT" position. Plug the DC power cord into the TV and insert the other end into the cigarette lighter on an auto. Depress the Power button and operate the TV in the same manner as when using AC power.

Service

Except for the batteries, there are no user serviceable parts in the battery pack. Refer any service to qualified service personnel.

RCA BLACK AND WHITE TELEVISION LIMITED WARRANTY

RCA Corporation guarantees to the first retail purchaser that if anything goes wrong with your portable RCA set within 90 days from the day you buy it—and it's our fault—we will pay your local RCA Authorized Servicenter its regular labor charges to repair it.

Additionally, we will make available new or, at our option, rebuilt replacements for defective parts, including the picture tube, for one year from the date of purchase. After the 90-day labor warranty expires, labor charges for installing replacement parts are your responsibility.

To obtain warranty repairs, you must take your set to an RCA Authorized Servicenter, present the warranty registration card, and pick up the set after repair.

The warranty is not transferable and excludes set installation or set-up, external antenna systems, adjustment of customer controls, sets purchased or serviced outside the U.S.A., and damage due to misuse or neglect.

**RCA Corporation, Consumer Electronics
600 North Sherman Drive
Indianapolis, Indiana 46201**

Warr. No. 1421302-2

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Marca(s) Registrada(s)
Printed in Taiwan**

RCA

RCM State
Division

from
P.R. Wakefield

TO

LOCATION

DATE

CTH

TV Interface TV Chip

Samples to you
within 30 days of
your request

Production is
starting and will
be available for
your customers.

PRW
9/14/76

PS.

Don Carley putting TV game demo
together for Japan Elec. Show.

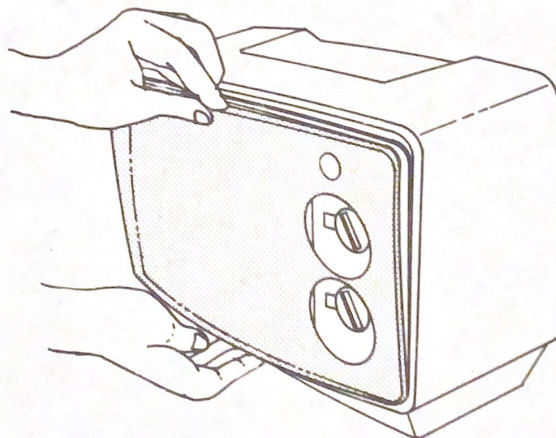
Light Filter Removal and Cleaning

To remove the light filter for cleaning, place your finger tips in the slot at the top center of the light filter as shown and pull straight out. With the top edge free but close to the TV, apply pressure at the bottom center of the light filter to lift and pull the filter away from the TV.

To replace the light filter, reverse the above steps, then push gently on the top corners. The filter will return to its original position.

To clean the filter, flush surface with cool water to remove the dust. Then wash in a solution of one cap full of liquid dishwashing detergent to a gallon of lukewarm water. Wash and dry with a clean soft cloth.

CAUTION—Do not attempt to clean the filter with a dry cloth as the dust will act as an abrasive and scratch the surface. Avoid the use of strong cleaning solutions, ammonia, or abrasive cleaners.





Earphone Operation

An earphone attachment is supplied with this model making possible personal, private listening.

Plug the earphone into the jack located on the front or side of the television cabinet.

When the earphone plug is inserted into the jack, the internal speaker is automatically disconnected.

Earphone Stock No. 15A137

1413556-1

AW326

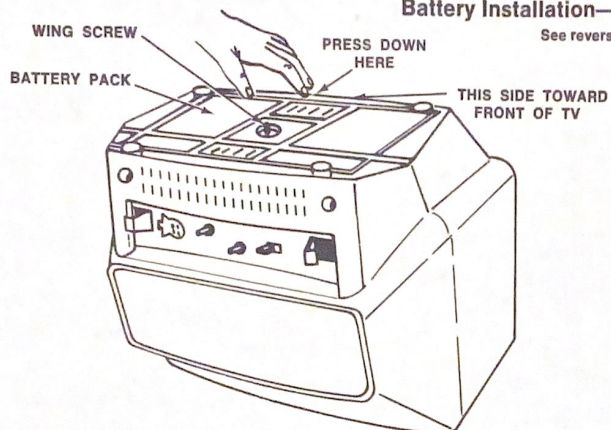
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Printed in Taiwan

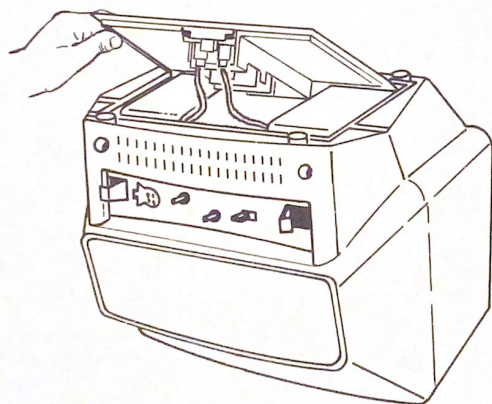
Battery Installation—Battery pack BP1200 (12 inch TV)

See reverse side for BP0900 (9 inch TV)



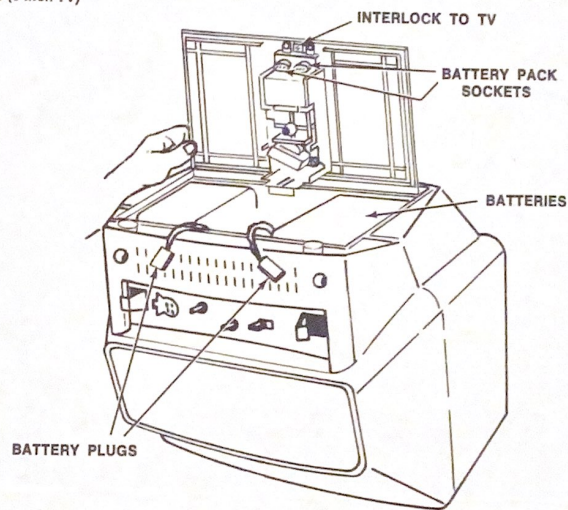
Step 1.

Turn TV over as shown. Apply pressure to front edge of pack and turn wing screw counterclockwise five complete turns. Remove battery pack.



Step 3.

Hold the pack over the TV and attach the battery plugs to the battery pack sockets. Bend the wires toward the inside and along side the batteries as shown. Set the front edge of the battery pack on the TV as shown and close the pack.



Step 2.

Place batteries in compartment on bottom of TV.

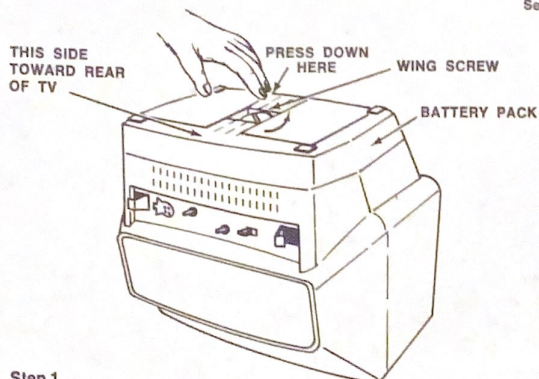
Step 4.

Turn wing screw clockwise until pack is firmly seated and wing screw is finger tight. Turn TV over. **Special Note:** Charge new batteries for 40-48 hours before using. This initial charge is necessary for maximum viewing time and battery life. See "Charging Batteries" in instruction book.

See **Battery Disposal Instructions** printed on battery label.

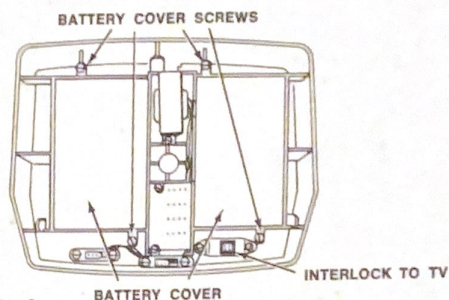
Battery Installation—Battery Pack BP0900 (9 inch TV)

See reverse side for BP1200 (12 inch TV)



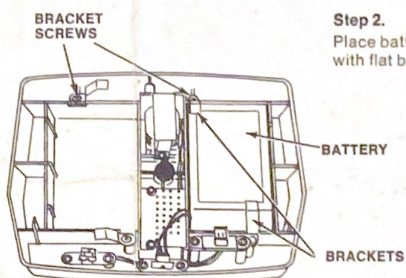
Step 1.

Turn TV over as shown. Apply pressure to front edge of pack and turn wing screw counterclockwise five complete turns. Remove battery pack.



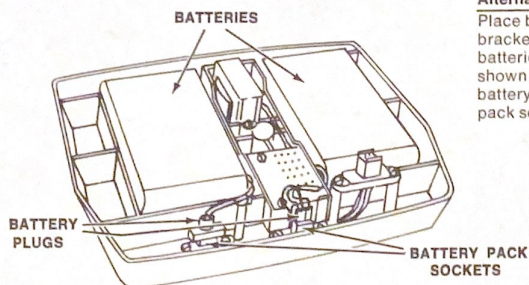
Step 2.

Place battery pack on table. Remove screws holding covers with flat blade screwdriver. Remove covers.



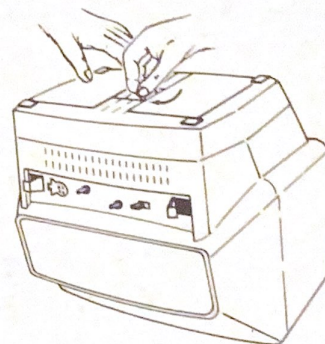
Alternate Steps 2 and 3 (used on some models).

Place battery pack on table. Loosen screws holding four (4) brackets and position brackets as shown at left. Place batteries in the pack, reposition brackets over batteries shown at right, and tighten screws. Attach battery plugs to battery sockets. Coil excess battery wire between battery pack sockets.



Step 3.

Place the batteries in the pack as shown. Attach battery plugs to battery sockets. Replace battery covers and secure with screws. Coil excess battery wire between battery pack sockets.



Step 4.

Set pack on TV. Turn wing screw clockwise until pack is firmly seated and wing screw is finger tight. Turn TV over.

Special Note: Charge new batteries for 40-48 hours before using. This initial charge is necessary for maximum viewing time and battery life. See "Charging Batteries" in instruction book.

See **Battery Disposal Instructions** printed on battery label.

~~ROSS~~
Paul Wahfield

DESCRIPTION OF COSMAC TV INTERFACE CHIP

FEATURES

Composite horizontal and vertical sync generation.
Schmitt trigger clear signal.
Programmable vertical resolution.
CDP1800 series compatible.
APAR C²L circuitry.
Interfaces directly with CDP1802 microprocessor
requiring only two pull up resistors.
Single voltage supply (4-6 volts).
External signals to enable and disable interrupts
and Direct Memory Access requests.
Clock frequency for 8-bit shift register can directly
connect to XTAL of CDP1802 for maximum horizontal
resolution.
1.76064 MHz processor clock frequency is required.
A 262 count horizontal line counter.
Fast COS/MOS 8-bit parallel to serial static shift register.
Restricted use of 3-byte instructions (long branches).
Approximately 50% of CPU processing time is dedicated to the
display.
60 Hz real time interrupt generator.

The RCA is a COS/MOS TV interface device designed to be connected directly to the CDP1802 COSMAC microprocessor. This device generates a composite sync signal and a luminence signal which can be combined externally to create a composite video signal. The composite video creates a non-interlaced, 262 lines per frame video picture. The display is a memory bit map with a maximum of 64 x 128 line segments. The vertical resolution is program controlled. The interrupt routine controls the memory addressing and the vertical resolution in synchronism with the direct memory access process to the display. Modifying the horizontal resolution requires additional hardware. Signals for enabling or disabling interrupt and direct memory access (DMA) requests are provided along with a general reset (RUN IN).

A typical CDP1802 microprocessor system using the TV interface is shown in Figure 1. All signals are compatible with the CDP1802 microprocessor. The system designer is free to choose the I/O flag and the I/O commands used for this device.

The is a single 24-lead hermetic dual-in-line ceramic package with the terminal assignment shown in Figure 2. The recommended operating range is 4-6 volts.

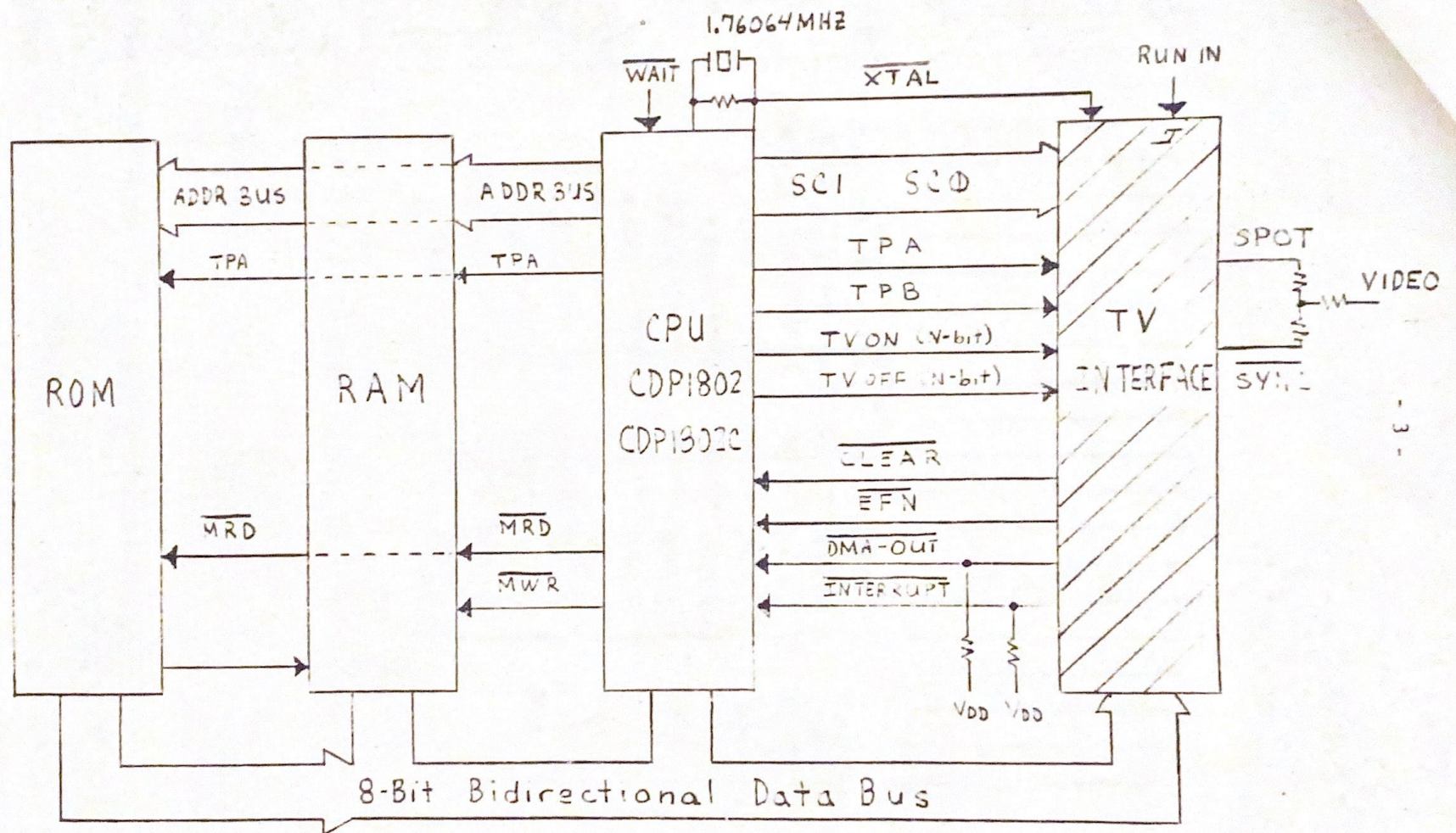
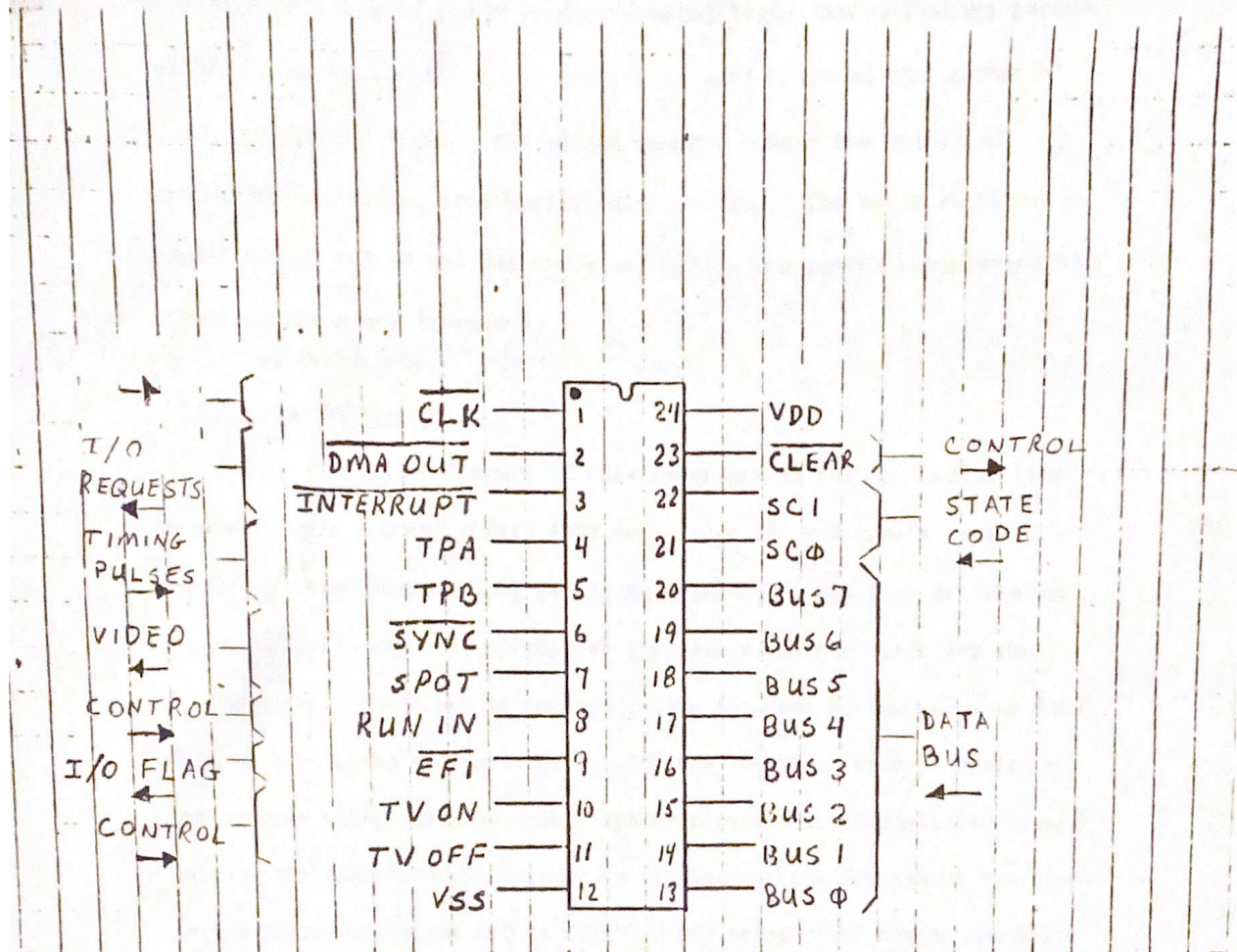


FIGURE 1 TYPICAL CDPI302 MICROPROCESSOR SYSTEM



TOP VIEW

TERMINAL ASSIGNMENT for _____,
the COSMAC TV INTERFACE CHIP

FIGURE 2.

ARCHITECTURE

The TV interface block diagram is shown in Figure 3. This interface consists of two counters, control logic, and a 8-stage parallel shift register. The 9-bit counter is used to count the number of lines per picture frame. The second counter counts the number of machine cycles during each horizontal scan line. The shift register is loaded during TPB of the DMA cycle and shifts the luminence information (SPOT) at the clock frequency.

a. Counters

A major feature of this interface is the horizontal line counter. This counter counts from zero up to 261 and resets to zero, repeating this cycle as long as timing pulses (TPA and TPB) are present. A machine-cycle counter, clocked by TPA, generates the clock for the line counter. There are 14 machine cycles for each horizontal sync pulse. This machine-cycle is synchronized with the fetch and execute cycles of the program being executed. This synchronization is accomplished by pre-setting the machine-cycle counter to 13 whenever the horizontal synchronization pulse occurs and SCO is "high". SCO being "low" during the horizontal synchronization pulse presets the counter to 12.

b. Control

A high pulse or level at the TVON input enables INTERRUPT and DMA requests, if RUN IN is high. The state TVON = 1 and RUN IN = 0 occurring simultaneously is not valid. INTERRUPT and DMA requests are disabled on the low-to-high transition of the TVOFF signal when TVON is low. Separate N-bits of the microprocessor can be used for TVON and TVOFF with no additional decoding.

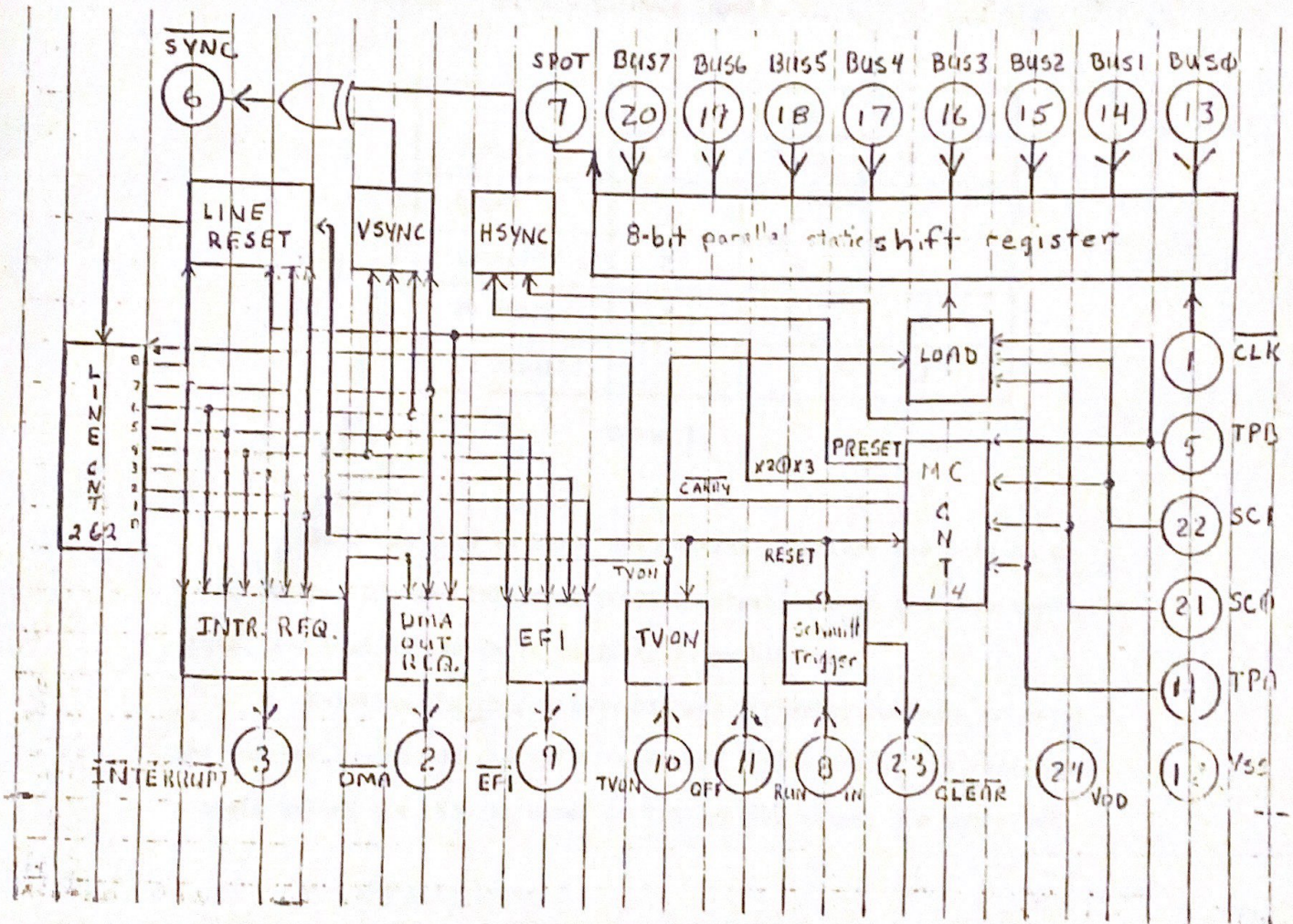


FIGURE 3. - TV INTERFACE CHIP BLOCK DIAGRAM

The RUN IN input is used as a general reset (RUN IN = 0). It has a Schmitt trigger-type input to eliminate oscillations at the $\overline{\text{CLEAR}}$ output when RUN IN is a slowly rising signal. Thus, a power-up reset can be accomplished with a RC network at this input.

STATE	SIGNAL		
	RUN IN	TV ON	TV OFF
RESET	L	L	X
INVALID	L	H	X
TV ENABLE	H	H	X
TV DISABLE	H	L	

Table 1.

When the chip is reset all internal counters are held in a preset state, DMA and INTERRUPT REQUEST outputs are in the high impedance state, and loading the shift register is prohibited.

Enabling the chip allows DMA and INTERRUPT requests to occur and permits loading of the shift register. $\overline{\text{EFI}}$ and $\overline{\text{SYNC}}$ are continuous signals unless the chip is reset or TPA and TPB pulses are inhibited.

c. Shift Register

An eight-stage parallel-input static shift register is clocked on the high-to-low transition of $\overline{\text{CLK}}$. As long as this device is enabled, the state TPB = 1, SCl = 1, and SCO = 0 permits the data bus to be loaded into the shift register at each high-to-low transition of $\overline{\text{CLK}}$ ¹.

1. At 5V the CDP1802 has a clock to TPB internal delay of typically 260 nsec.

SPOT is the most significant bit (MSB) of the shift register. Whenever the shift register is not being loaded, a "0" is shifted into the least significant bit (LSB) at each high-to-low transition of $\overline{\text{CLK}}$.

SIGNAL DESCRIPTIONS

TPA, TPB

Positive timing pulses from the microprocessor which occur once in each machine cycle. TPB follows TPA (see Figure 5a). Each machine cycle should be 4.5438 microseconds. These signals are used to clock internal counters and to time interaction with the data bus.

$\overline{\text{CLK}}$

Normally connected to $\overline{\text{XTAL}}$ of the CDPL802. This signal is the clock frequency of the shift register. The high-to-low transition of $\overline{\text{CLK}}$ shifts the 8-bit register left (the most significant bit out first). "0" is shifted into the least significant bit. When the chip is enabled, data is loaded into the shift register on the high-to-low transition of CLK with TPB = 1, SCL = 1, and SCO = 0. The clock frequency (1.76064 MC) equals the number of frames per second (60), times the number of lines per frame (262), times the number of machine cycles per line (14), times the number of bits per byte (8).

BUS 0-7

Eight-bit bidirectional DATA BUS lines. These signals are the inputs to the 8-stage shift register.

DMA OUT

Requests picture information from the microprocessor when low. An external pull-up resistor to V_{DD} is required for this signal line. DMA OUT requests occur during the 65th through 193rd horizontal lines, following the vertical sync pulse. See Figure 4. Each request is eight machine cycles long (Figure 5a). No DMA OUT requests are generated if the chip is disabled or reset (see Table 1).

INTERRUPT

Held low for two periods of the horizontal frequency before the first horizontal display line (see Figures 4 and 5b). Interrupt requests are inhibited when the interface chip is disabled or reset (see Table 1). This signal line requires an external pull up resistor to V_{DD} .

EFl

Low for four periods of the horizontal frequency before both the beginning and end of the Display Area. Figure 4 shows the relationship between this signal and the video picture frame. It is used by the interrupt routine to determine the end of the display when not displaying the maximum amount (1024 bytes) of display memory. This signal may also be used to synchronize the main program with the interrupts. This signal line is not a tri-state output.

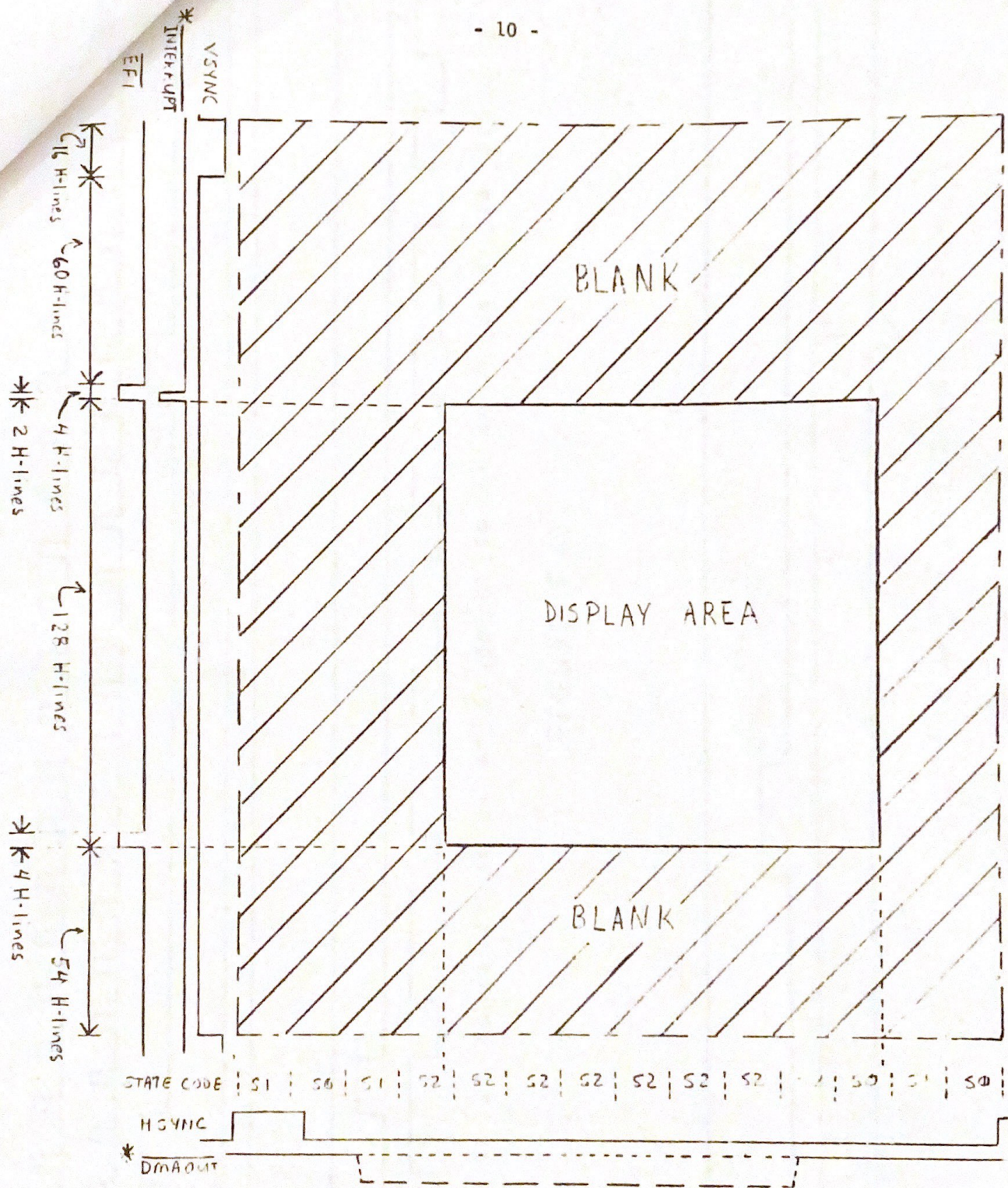


FIGURE 4 SPATIAL DIAGRAM OF ONE PICTURE FRAME
(NOT TO SCALE)

* TRI-STATE OUTPUTS REQUIRE EXTERNAL PULL UP RESISTORS TO V_{DD}

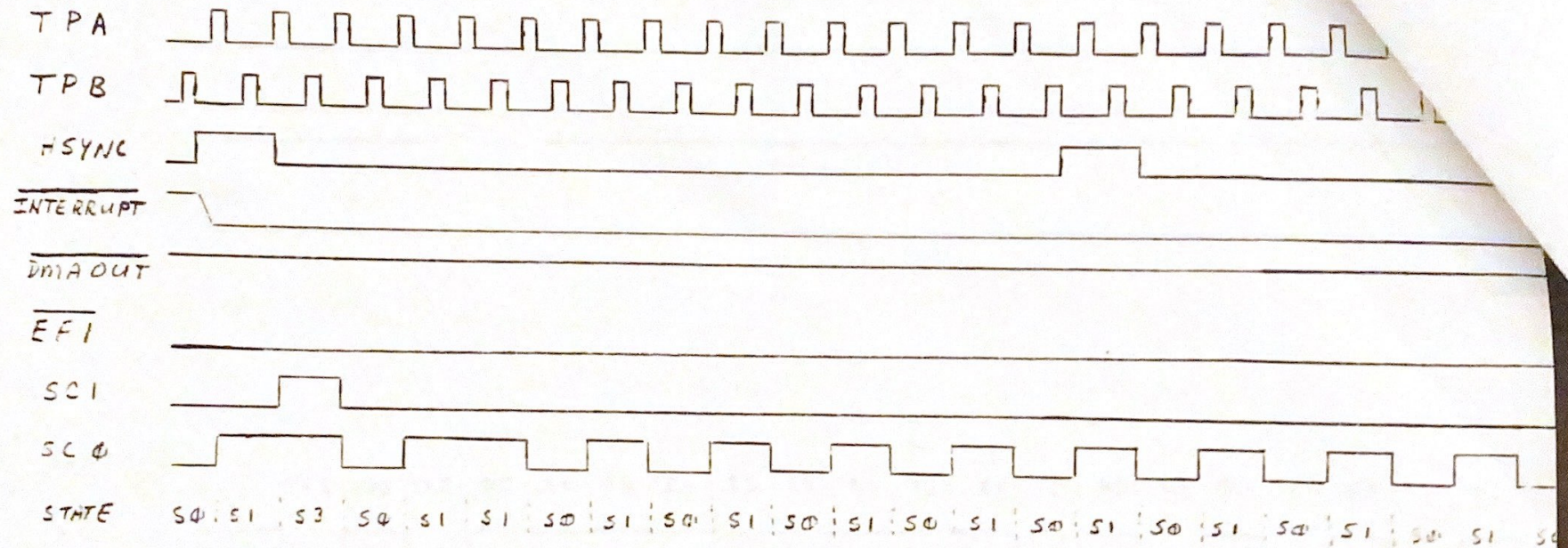


FIGURE 5 a)

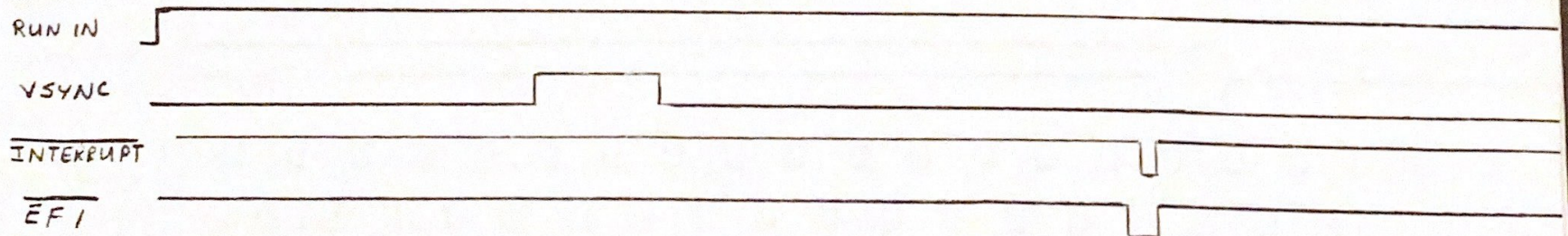


FIGURE 5.b) TIMING DIAGRAM

Continued
on next
page

SCO, SC1

Two state code signals of the CDP1802. They determine when data is loaded into the shift register. SCO is also used for presetting an internal counter to be synchronous with the program being executed.

SPOT

Most significant bit output of the 8-bit shift register. This signal is used to determine the luminence level for the composite video signal. The composite video signal may be obtained by resistively coupling SPOT with $\overline{\text{SYNC}}$ (see Figure 1).

$\overline{\text{SYNC}}$

The Exclusive "OR" of the horizontal and the vertical SYNC signals. $\overline{\text{SYNC}}$ can be combined externally with SPOT to generate a composite video signal.

RUN IN

General reset for this device (RUN IN = 0). The internal counters are preset. Schmitt trigger type-input eliminates oscillations at $\overline{\text{CLEAR}}$ to allow for a slowly rising power-up reset signal. See Table 1 for reset conditions.

TVON, TVOFF

Signals from microprocessor to set and reset a flip-flop. The state of this flip-flop is used to enable or disable the processor interface signals $\overline{\text{DMA OUT}}$ and $\overline{\text{INTERRUPT}}$. Separate microprocessor N-bits may be used directly for these control inputs. See Table 1 for enable/disable conditions.

PROGRAMMING CONSIDERATIONS

Once the TV interface chip is electrically connected to the CDP1802 microprocessor there are several programming conventions that must be considered. An interrupt routine is necessary for properly controlling the information being sent to the display. For timing purposes one of the first five instructions executed in the interrupt routine must be a three cycle instruction. No other three cycle instruction should be executed while this interface is enabled. Before enabling the chip, Register 1 of the processor should be initialized to the beginning of the interrupt routine. R2 is the stack pointer. Before each interrupt, R2 must be pointing to the top of a stack with at least two available RAM locations. Note that in the sample interrupt routines, the stack is assumed to be used outside the interrupt routine and R2 is decremented to point to the next available RAM location. R0 is the DMA memory pointer. The above three registers should not be used for other functions while the TV interface is enabled. Figures 6, 7, 8 and 9 are sample INTERRUPT routines used to display 8 bytes, 256 bytes, 512 bytes, or 1024 bytes of memory. All "E2" instructions are used as no-ops (since $X = 2$ during entire interrupt routines) and may be replaced by any two machine cycle instruction, but the number of instructions executed from the beginning of the interrupt to the end of the display must remain the same as those in Figures 6, 7 and 8. Other functions may be implemented in the interrupt routine following the required display refresh functions and before returning to the previously executing program. Other formats for the display can be generated by variations in the interrupt routine, including vertical scrolling of the display.

In the sample interrupt routines, R0 is decremented to keep the DMA pointer within the display page when there are multiple lines of the same information. The "] " in the machine code indicates where the DMA's for the display should occur. The D register is used to restore the low order address of R0 to the beginning address of the current display line, whenever a display line is repeated.

Note that for Figure 9 the interrupt routine only has to initialize R0 to the starting address of the display but must last at least 28 machine cycles before returning to the main program in order to avoid multiple interrupts. Figure 4 illustrates that the INTERRUPT request signal remains low for two horizontal lines (28 machine cycles) before the top line of the display. Using the interrupt routine of Figure 9 to display 1024 bytes of memory requires less than thirty per cent of the CPU time for display refresh.